



Python Basics

What is Python?

Python is a high-level, interpreted programming language that is designed to be simple, readable, and easy to learn. It was created by Guido van Rossum and first released in 1991.



One of the reasons Python is so popular is because its syntax looks very close to normal English, making it beginner-friendly compared to many other programming languages.

Python is used in several areas of technology including web development, artificial intelligence, automation, data science, cybersecurity, and software development.

Why is Python Popular?



Python is popular because it is easy to read, easy to write, and very powerful.

Unlike some programming languages that require a lot of complex syntax, Python allows developers to focus more on solving problems rather than worrying about complicated code structures.

Another reason Python is widely used is because it has a massive community and thousands of libraries that help developers build applications faster.

Companies like Netflix, Instagram, and Spotify use Python in different parts of their systems.

USES OF PYTHON

Python is a versatile programming language used in a wide range of applications and industries.



WEB DEVELOPMENT

Python frameworks like Django and Flask help build powerful websites and web applications.



MACHINE LEARNING & AI

Python has powerful libraries like TensorFlow, PyTorch and Scikit-learn for building ML models.



DATA SCIENCE

Python is widely used for data analysis, visualization and insights extraction.



DATA VISUALIZATION

Libraries like Matplotlib, Seaborn and Plotly help create meaningful visualizations.



AUTOMATION

Python can automate repetitive tasks like file handling, web scraping, testing, etc.



CLOUD COMPUTING

Python is used in cloud platforms and tools to build scalable and robust applications.



CYBERSECURITY

Python is used for ethical hacking, security testing, network scanning and building security tools.



DESKTOP APPLICATIONS

Python can be used to build desktop GUI applications using Tkinter, PyQt, and other libraries.



GAME DEVELOPMENT

Python is beginner-friendly for game development using libraries like Pygame and Panda3D.



Python's simplicity, readability and huge community support make it one of the **most popular programming languages** in the world.

How Do You Install Python?



To start programming with Python, the first step is to install it on your computer.

You can download Python from the official website:

<https://www.python.org/>

After installation, you can write Python code using editors like:

Visual Studio Code

PyCharm

What is Python Syntax?



```
1  print("Hello, World")  
2  
3  # Output: Hello, World  
4
```

Syntax simply means the rules used to write code in a programming language.

Python syntax is very clean and beginner-friendly.

For example, to display text on the screen, we use the print function.

Quick Exercise



Write a Python Program that asks the user for two numbers and:

- 1.) Displays the Sum
- 2.) Displays the difference
- 3.) Displays the Product

What are Variables in Python?



Variables are used to store data in programming.

Variables make it easy for a particular value to be reused

Variables also make our code cleaner as we avoid unnecessary repetition

Examples on Variables

USING VARIABLES



```
1 a = 5
2 b = 10
3 sum = a + b
4 avg = sum / 2
5 print(f"Value For A is: ${a}")
6 print(f"Value For B is: ${b}")
7 print(f"Sum is: ${sum}")
8 print(f"Average is: ${avg}")
9
```

WITHOUT VARIABLES



```
1 print(f"Value For A is: ${5}")
2 print(f"Value For B is: ${10}")
3 print(f"Sum is: ${10+5}")
4 print(f"Average is: ${(10+5)/2}")
5
```

What are Data Types in Python?

Data types describe the type of data stored in a variable.

Some common data types in Python include:

String → Text values

Integer → Whole numbers

Float → Decimal numbers

Boolean → True or False values

List → Collection of items

Dictionary → Data stored in key-value pairs





```
1  name = "Favour"  # string
2
3  age = 23  # Integer
4
5  weight = 59.5  # float
6
7  is_active = True  # boolean
8
9  fruits = ["Mango", "Orange", "Pine-apple"]  # list
10
11  favour = {
12      "lname": "Tony-Akinlosotu",
13      "fname": "Favour",
14      "oname": "Adeniran",
15      "age": 24,
16  }  # dictionary
17
```



ARITHMETIC OPERATORS



Arithmetic operators are used to perform mathematical operations.

OPERATOR	NAME	DESCRIPTION	EXAMPLE	RESULT
	Addition	Adds two values	$5 + 3$	
	Subtraction	Subtracts the second value from the first	$5 - 3$	
	Multiplication	Multiplies two values	$5 * 3$	
	Division	Divides the first value by the second	$5 / 3$	
	Floor Division	Divides and returns the largest integer less than or equal to the result	$5 // 3$	
	Modulus	Returns the remainder of division	$5 \% 3$	
	Exponentiation	Raises the first value to the power of the second	$5 ** 3$	



WHAT ARE CONTROL STATEMENTS IN PYTHON?





Control Statements

Control Statements are blocks are constructs that are used to alter the normal or sequential flow of a program.

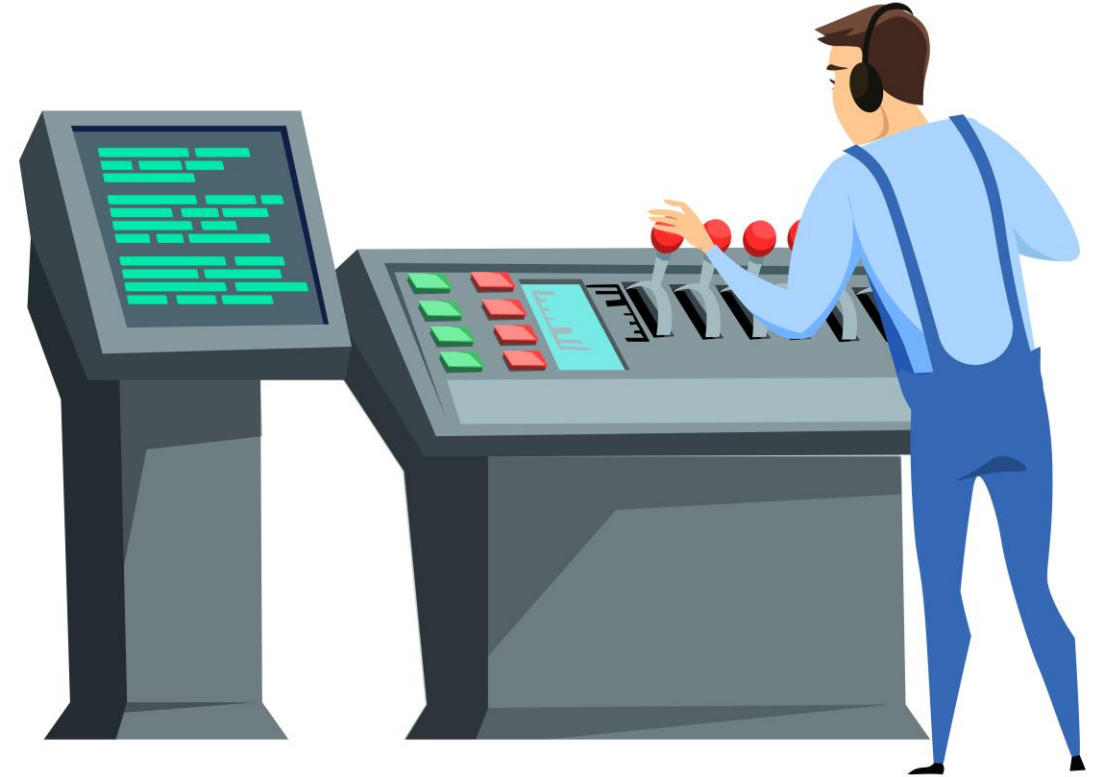
It allows the program to make decisions or repeat actions depending on the fulfillment of the defined condition





Types of Control Statements

- Conditional Control Statement
- Iterative Control Statement





Conditional Control Statements

In Python we use:

- If
- elif
- else

```
1  number = 10
2
3  if number > 0:
4      print("Number is Positive")
5  elif number < 0:
6      print("Number is Negative")
7  else:
8      print("Number is Zero (0)")
9
```


Quick Exercise



Write a Python Program that asks the user for two numbers and:

- 1.) Decides if one is greater than the other and which one is greater OR
- 2.) Decides if they are equal
- 3.) Prints the result

Take Home



Create a Python file and name it, “**modulus.py**”

Inside the python file, write a program a number and decides if the number is an **even** or **odd** number



Iterative Control Statements

Python mainly uses:

- For-loop
- While loop



```
1  # A program that prints from 1 to 5
2  print("For Loop")
3  for i in range(1, 6):
4      print(i)
5
6
7  print("While Loop")
8  i = 1
9  while i < 6:
10     print(i)
11     i = i + 1
12
```



PYTHON LIBRARIES & FRAMEWORKS

Python has a rich ecosystem of libraries and frameworks that make development faster, easier and more powerful.

POPULAR LIBRARIES



NumPy

For numerical computing and working with arrays and matrices.



pandas

For data analysis and manipulation using DataFrames.



Matplotlib

For creating static, animated and interactive visualizations.



Seaborn

For statistical data visualization built on top of Matplotlib.

Requests

For making HTTP requests easily and efficiently.

Beautiful soup

For web scraping and parsing HTML / XML.

POPULAR FRAMEWORKS

django

A high-level web framework that encourages rapid development and clean, pragmatic design.

Web Development



Flask

A lightweight and flexible micro web framework for building web applications and APIs.

Web Development



FastAPI

A modern, fast (high-performance) web framework for building APIs with Python 3.7+ based on standard type hints.

Web Development (APIs)



TensorFlow

An open-source library for Machine Learning and Deep Learning.

Machine Learning



PyTorch

An open-source Machine Learning library based on Torch, used for Deep Learning applications.

Machine Learning



These libraries and frameworks help developers avoid reinventing the wheel and focus on building **powerful, scalable and efficient applications.**



**Why Should
Beginners Learn
Python?**



Reasons to Learn Python

- ❑ It is easy to understand
- ❑ It has simple syntax
- ❑ It is highly in demand
- ❑ It has many career opportunities



Learning Python can open doors to careers in software engineering, data science, artificial intelligence, automation, and cybersecurity.



Thanks

